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Matter Smart Relay with the Arduino Nano Matter

Learn how to build a Matter smart relay to control an AC lightbulb using the Arduino Nano Matter

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Overview

This tutorial will teach you how to use the Arduino Nano Matter to create a Matter smart relay to control an AC lightbulb.



Smart relay overview

Thanks to the seamless compatibility of the Nano Matter with almost any Matter network we can easily integrate our smart relay with Amazon Alexa, Google Assistant, Apple Home, Home Assistant and even any professional custom solution.

We have prepared a short demo in video format in case you are a visual learner.

[Help](#)

Build a Matter-Enabled
Lightbulb with the Arduino
Nano Matter



Hardware and Software Requirements

Hardware Requirements

[Arduino Nano Matter](#) (x1)

Grove Relay Module (x1)

Lightbulb socket (x1)

AC lightbulb (x1)

AC cable (x1)

Breadboard (x1)

Jumper wires

Google Nest Hub Max (Thread
Border Router) (x1)

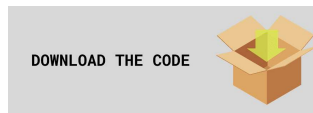
[USB-C® cable](#) (x1)

Software Requirements

[Arduino IDE 2.0+](#) or [Arduino](#)

— — — —

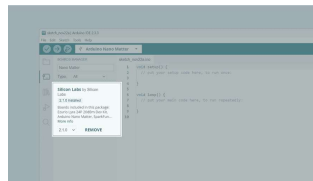
Download the Project Code



Download the complete project code [here](#).

Board Core and Libraries

The **Silicon Labs** core contains the libraries and examples you need to work with the board's components, such as its Matter, Bluetooth® Low Energy, and I/Os. To install the Nano Matter core, navigate to **Tools > Board > Boards Manager** or click the Boards Manager icon in the left tab of the IDE. In the Boards Manager tab, search for `Nano Matter` and install the latest `Silicon Labs` core version.

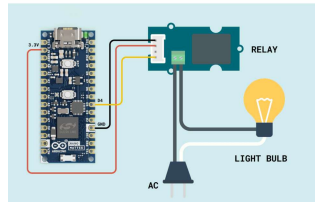


Installing the Silicon Labs core in the Arduino IDE

Project Setup

Schematic Diagram

Use the following connection diagram for the project:



Project wiring diagram

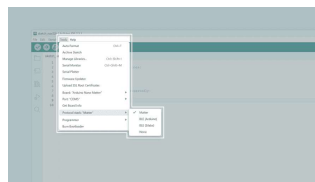
The relay module is powered by the Nano Matter `3.3V` output pin and the relay is controlled with the `D4` pin. Wire the AC cable with the lightbulb so the relay interrupts the current path when open.



Be careful when handling with main voltages.

Programming

In the Arduino IDE upper menu, navigate to **Tools > Protocol stack** and select **Matter**.



Matter Protocol stack
selected

Copy and paste the following sketch:

```

1  #include <Matter.h>
2  #include <MatterLightbulb.h>
3
4  #define RELAY D4 // Relay pin
5
6  MatterLightbulb matter_bulb;
7
8  void setup()
9  {
10     Serial.begin(115200);
11     Matter.begin();
12     matter_bulb.begin();
13
14     pinMode(BTN_BUILTIN, INPUT);
15
16     pinMode(LED_BUILTIN, OUTPUT);
17     digitalWrite(LED_BUILTIN, LOW);
18
19     pinMode(RELAY, OUTPUT);
20     digitalWrite(RELAY, LOW);
21
22     Serial.println("Matter connected");
23
24     if (!Matter.isDeviceConnected())
25     {
26         Serial.println("Matter device not connected");
27         Serial.println("Connect the device to the network");
28         Serial.printf("Manual connection: %s", IP_ADDRESS);
29         Serial.printf("QR code: %s", QR_CODE);
30     }
31 }

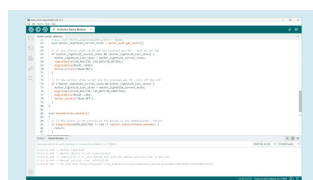
```

The structure of this example code is very simple, the main functions are explained below:

In the `setup()` function we initialize the Matter connectivity and the relay output pin.

In the `loop()` function we listen to any relay toggle command sent from the smartphone or voice assistant.

Once you uploaded the example code to the Nano Matter, open the Serial Monitor and reset the board.

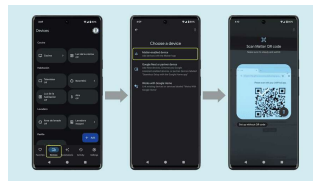


After the reset you will find on the serial port the URL that generates the QR for the Matter device commissioning.

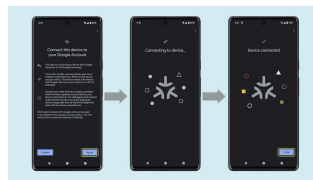
Adding the Device (Commissioning)

Copy and paste the QR code URL on your favorite web browser and a unique QR code will be generated for your board.

Go to your **Google Home** app, navigate to **devices** and tap on **Add**, select the **Matter-enabled device** option and scan the QR code.



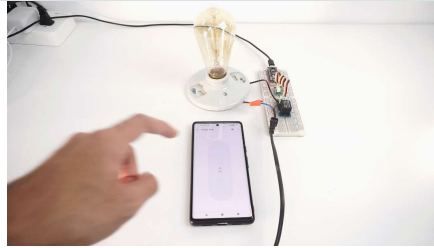
Adding the device to
Google Home app



Your device is successfully
added

Final Results

Finally, you will be able to control the AC lightbulb from your smartphone, hub or asking your personal assistant.



Conclusion

In this tutorial we have learned how to create a Matter enabled smart relay that can be controlled from our smartphone and personal assistant. The Nano Matter allows us to seamlessly integrate our own smart relay as a commercial product with our current smart home ecosystem.

Next Steps

You can take this solution even further by adding several relay channels or solid state relays to control more than one load.

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